From leaf images to a reproducible comparison

PlantVillage

Tomato images
18146 cleaned

Grouped split

Train 12701
Val 2723
Test 2722

Fixed baseline

Preregistered recipe

Recipe search

DE + random search
Validation macro-F1

Rerank recipes

Higher fidelity
Validation only

Final training

Same epoch budget
3 training seeds
Fixed recipes

Held-out evaluation

Metrics + CIs
Paired tests
Corruptions + Grad-CAM

Evidence + demo

JSON → figures / report
Exact checkpoints
Offline diagnosis

Test images never guide search, reranking, early stopping or model selection.